Mathematical and Computational Methods

Lecture 6: Eigensystems

This lecture uncovers the deepest structure a square matrix carries: its *eigenvalues* and *eigenvectors*. A matrix transformation usually sends a vector to a new length *and* a new direction, but along a few special directions it does something far simpler—it merely stretches, leaving the direction unchanged. Those directions are the *eigenvectors*, the stretch factors are the *eigenvalues*, and together they are the natural axes of the matrix. We find them from a determinant equation, use them to *diagonalise* a matrix and so compute its powers cheaply, and finally watch them govern the long-run behaviour of a system evolving in steps—a Markov chain.

Where this fits. Eigensystems gather up the whole linear-algebra strand. The eigenvalues are roots of a *determinant* equation, each eigenvector is found by solving a homogeneous system with *Gaussian elimination*, and diagonalisation is built from an *inverse*—all three from Lecture 5; the eigenvectors are the invariant directions of its transformations, drawn with the vectors of Lecture 4, and an eigenvalue may even be *complex*, the numbers of Lecture 3. This is where the linear-algebra strand ends, but not where the ideas do: mathematics and data-science students meet eigenvalues again in a dedicated linear-algebra course—they underlie principal-component analysis and much of modern data science—while physicists meet them as the normal modes of a vibrating system and the observables of quantum mechanics.

Contents

Learning objectives

By the end of this lecture you will be able to:

- Find eigenvalues and eigenvectors of 2×2 and 3×3 matrices, with a trace/determinant check.
- Interpret complex eigenvalues as rotation, and the eigenstructure of symmetric and Hermitian matrices.
- Decide diagonalisability and use diagonalisation to compute matrix powers.
- Apply orthogonal diagonalisation and analyse a Markov chain via its dominant eigenvalue.

1 Eigenvalues and eigenvectors

1.1 The idea: directions a matrix preserves

A transformation generally turns a vector to a new direction. But for most matrices there are a few special directions that are *not* turned—along them the matrix only stretches or shrinks. Such a direction is an *eigenvector*, and its stretch factor is the *eigenvalue*. A pure stretch leaves the coordinate axes pointing as they were, a reflection fixes its mirror line, and a plane rotation—as we shall see—fixes no real direction at all.

Definition 1.1 (Eigenvalue and eigenvector). Let $\mathbf{A}$ be a square matrix. A *non-zero* vector $\mathbf{v}$ is an *eigenvector* of $\mathbf{A}$, with *eigenvalue* λ, if

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

The eigenvector marks a direction the transformation preserves; the eigenvalue is the factor by which it stretches ($\lambda > 1$), shrinks ($0 < \lambda < 1$), or flips and scales ($\lambda < 0$) along that direction. The zero vector is excluded, since $\mathbf{A}\mathbf{0} = \lambda\mathbf{0}$ holds for every λ and says nothing.

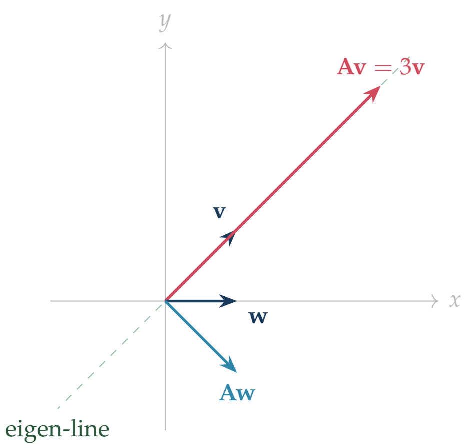

Figure 1: For $\mathbf{A} = \left(\begin{smallmatrix} 1 & 2 \\ -1 & 4 \end{smallmatrix}\right)$ with eigenvector $\mathbf{v} = (1, 1)$ and eigenvalue 3: the eigenvector $\mathbf{v}$ stays on its line ($\mathbf{A}\mathbf{v} = 3\mathbf{v}$ lies on the dashed line), while the ordinary vector $\mathbf{w}$ is swung to a new direction.

Figure 1 shows the matrix of the running example below acting on one of its eigenvectors and on an ordinary vector. Eigenvectors expose the skeleton of a transformation: once we know the directions it merely scales, and by how much, we know almost everything about how it acts.

1.2 Finding eigenvalues and eigenvectors

To *find* eigenvalues we turn $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ into a condition with no $\mathbf{v}$ in it. Writing $\lambda\mathbf{v} = \lambda\mathbf{I}\mathbf{v}$ and collecting terms,

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \iff (\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}.$$

This homogeneous system has a *non-zero* solution $\mathbf{v}$ precisely when its matrix is singular—when, by Lecture 5, its determinant vanishes.

Definition 1.2 (Characteristic equation). The eigenvalues of $\mathbf{A}$ are the solutions of the *characteristic equation*

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0.$$

The left-hand side is the *characteristic polynomial*, of degree n for an $n \times n$ matrix; its roots are the eigenvalues. By the fundamental theorem of algebra (Lecture 3) there are always n such roots in $\mathbb{C}$, counted with multiplicity—though some may be complex or repeated. For each eigenvalue, the eigenvectors are the non-zero solutions of $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$, found by the elimination of Lecture 5.

Example 1.1 ***(Eigenvalues and eigenvectors of a 2×2 matrix).*** Take $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$, the matrix of Figure 1. The characteristic equation is

$$\det\begin{pmatrix} 1-\lambda & 2 \\ -1 & 4-\lambda \end{pmatrix} = (1-\lambda)(4-\lambda) - (2)(-1) = \lambda^2 - 5\lambda + 6 = (\lambda - 2)(\lambda - 3) = 0,$$

so the eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 3$. For $\lambda_1 = 2$, solve $(\mathbf{A} - 2\mathbf{I})\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix}\mathbf{v} = \mathbf{0} \implies -v_1 + 2v_2 = 0 \implies v_1 = 2v_2,$$

giving $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ (and any multiple). For $\lambda_2 = 3$, $(\mathbf{A} - 3\mathbf{I})\mathbf{v} = \mathbf{0}$ reads $\begin{pmatrix} -2 & 2 \\ -1 & 1 \end{pmatrix}\mathbf{v} = \mathbf{0}$, i.e. $v_1 = v_2$, so $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. As a check, $\mathbf{A}\mathbf{v}_2 = (3, 3)^\mathsf{T} = 3\mathbf{v}_2$. These are the two eigen-directions of the figure; note they are *not* perpendicular.

The method is identical in three dimensions, only the determinant is larger.

Example 1.2 ***(Eigenvalues and eigenvectors of a 3×3 matrix).*** Take $\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}$. Expanding the characteristic determinant along the first row (the corner zero keeps it short),

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (2-\lambda)\big[(3-\lambda)(2-\lambda) - 1\big] - 1 \cdot (2-\lambda) = (2-\lambda)\big[(3-\lambda)(2-\lambda) - 2\big].$$

Since $(3-\lambda)(2-\lambda) - 2 = \lambda^2 - 5\lambda + 4 = (\lambda - 1)(\lambda - 4)$, the characteristic equation factors as $(2-\lambda)(\lambda-1)(\lambda-4)=0$, so $\lambda = 1, 2, 4$. Take $\lambda = 1$ in detail and row-reduce $\mathbf{A} - \mathbf{I}$:

$$\mathbf{A}-\mathbf{I} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_2 \to R_2 - R_1} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

The zero row leaves one free unknown; setting $v_3 = 1$, back-substitution gives $v_2 = -1$ and $v_1 = 1$, the eigenvector $(1, -1, 1)$. The same elimination on $\mathbf{A} - 2\mathbf{I}$ and $\mathbf{A} - 4\mathbf{I}$ produces the others, so

$$\lambda = 1: \ \mathbf{v} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \quad \lambda = 2: \ \mathbf{v} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \lambda = 4: \ \mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

A check confirms each: for $\lambda = 4$, $\mathbf{A}(1,2,1)^\mathsf{T} = (4,8,4)^\mathsf{T} = 4(1,2,1)^\mathsf{T}$. The three eigenvalues here are *distinct*, and the eigenvectors turn out mutually orthogonal—no accident, as Section 2.3 explains. (Eigenvalues that *repeat* need a little more care; that case is taken up in Section 2.4.)

Note (Two quick checks: trace and determinant). The trace and determinant of Lecture 5 give a free check on any eigenvalue computation: the eigenvalues of an $n \times n$ matrix always satisfy

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = \operatorname{tr} \mathbf{A}, \qquad \lambda_1 \lambda_2 \cdots \lambda_n = \det \mathbf{A}.$$

For the 2×2 example $\operatorname{tr} \mathbf{A} = 1 + 4 = 5 = 2 + 3$ and $\det \mathbf{A} = 6 = 2 \cdot 3$; for the 3×3, $\operatorname{tr} \mathbf{A} = 2 + 3 + 2 = 7 = 1 + 2 + 4$ and $\det \mathbf{A} = 8 = 1 \cdot 2 \cdot 4$. A mismatch is a sure sign of an arithmetic slip.

The note states these two checks; both fall straight out of the characteristic polynomial, and the 2×2 case is worth seeing in full.

Proposition 1.1 (Trace and determinant from the eigenvalues). Let $\mathbf{A}$ be an $n \times n$ matrix with eigenvalues $\lambda_1, \dots, \lambda_n$—the roots of its characteristic polynomial, counted with multiplicity. Then

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = \operatorname{tr} \mathbf{A}, \qquad \lambda_1 \lambda_2 \cdots \lambda_n = \det \mathbf{A}.$$

Proof. The eigenvalues are by definition the roots of $\det(\mathbf{A} - \lambda \mathbf{I})$, so this polynomial is determined by them. Take the 2×2 case first, with $\mathbf{A} = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Expanding the determinant,

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (a-\lambda)(d-\lambda) - bc = \lambda^2 - (a+d)\lambda + (ad - bc) = \lambda^2 - (\operatorname{tr}\mathbf{A})\lambda + \det \mathbf{A}.$$

The very same polynomial, written from its roots λ_1, λ_2, is

$$(\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2.$$

Matching the coefficients of λ and of the constant term gives $\lambda_1 + \lambda_2 = \operatorname{tr}\mathbf{A}$ and $\lambda_1\lambda_2 = \det\mathbf{A}$ at once.

For general n the same coefficient-matching works on the monic characteristic polynomial $\chi(\lambda) := \det(\lambda\mathbf{I} - \mathbf{A}) = (-1)^n \det(\mathbf{A} - \lambda\mathbf{I})$, which has the same roots. Written from those roots,

$$\chi(\lambda) = \prod_{i=1}^{n} (\lambda - \lambda_i) = \lambda^n - \left(\textstyle\sum_i \lambda_i\right)\lambda^{n-1} + \cdots + (-1)^n \textstyle\prod_i \lambda_i.$$

Expanding $\det(\lambda\mathbf{I}-\mathbf{A})$ instead, the term in λ^{n-1} can come only from the diagonal product $(\lambda - a_{11})\cdots(\lambda - a_{nn})$—any off-diagonal factor drops two powers of λ at once—and its coefficient is $-(a_{11}+\cdots+a_{nn}) = -\operatorname{tr}\mathbf{A}$, while the constant term is $\chi(0) = \det(-\mathbf{A}) = (-1)^n\det\mathbf{A}$. Comparing the two expansions gives $\sum_i \lambda_i = \operatorname{tr}\mathbf{A}$ and $(-1)^n\prod_i\lambda_i = (-1)^n\det\mathbf{A}$, that is, $\prod_i\lambda_i = \det\mathbf{A}$. ■

1.3 Complex eigenvalues

A real matrix need not have real eigenvalues—and geometrically the reason is rotation, which leaves no real direction fixed.

Example 1.3 *(A rotation has complex eigenvalues).* The quarter-turn $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ turns every vector by $90°$, so it should fix no real direction. Its characteristic equation confirms this:

$$\det\begin{pmatrix} -\lambda & -1 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 + 1 = 0 \implies \lambda = \pm\mathrm{i}.$$

The eigenvalues are the imaginary numbers $\pm\mathrm{i}$ of Lecture 3, with complex eigenvectors $(1,-\mathrm{i})^{\mathsf{T}}$ for $\lambda = \mathrm{i}$ and $(1,\mathrm{i})^{\mathsf{T}}$ for $\lambda = -\mathrm{i}$. There are no real eigenvectors, exactly as the geometry demands.

More generally the rotation by angle θ, $\mathbf{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, has characteristic equation $\lambda^2 - 2\cos\theta\,\lambda + 1 = 0$ and hence eigenvalues

$$\lambda = \cos\theta \pm \mathrm{i}\sin\theta = \mathrm{e}^{\pm\mathrm{i}\theta},$$

the unit-modulus complex numbers of Lecture 3. The modulus 1 records that a rotation preserves length, and the argument θ is the angle turned. Complex eigenvalues are the algebraic signature of rotation; whenever a system oscillates or spirals, complex eigenvalues are at work, their modulus setting the growth or decay and their argument the frequency.

1.4 Symmetric and Hermitian matrices

Two families of matrices have especially well-behaved eigensystems, and both are central to applications. A real *symmetric* matrix ($\mathbf{A}^{\mathsf{T}} = \mathbf{A}$) has eigenvalues that are all *real* and eigenvectors that, for distinct eigenvalues, are *orthogonal*. The 3×3 matrix of Example 1.2 is symmetric, and indeed its eigenvalues $1, 2, 4$ came out real and its eigenvectors mutually orthogonal—no coincidence, but a guarantee.

The complex counterpart is a *Hermitian* matrix.

Definition 1.3 (Hermitian matrix). A complex square matrix is *Hermitian* if it equals its own conjugate transpose,

$$\mathbf{A}^{\dagger} = \mathbf{A}, \qquad \mathbf{A}^{\dagger} := \overline{\mathbf{A}}^{\mathsf{T}},$$

that is, transposing *and* conjugating every entry (Lecture 3) returns $\mathbf{A}$. The diagonal entries are therefore real, and entries across the diagonal are conjugates. Like symmetric matrices, Hermitian matrices have only real eigenvalues.

Example 1.4 (*A Hermitian* 2×2)**.** Let $\mathbf{H} = \begin{pmatrix} 2 & 1-\mathrm{i} \\ 1+\mathrm{i} & 3 \end{pmatrix}$. Conjugating and transposing swaps the off-diagonal entries and flips the sign of i, returning $\mathbf{H}$, so $\mathbf{H}^\dagger = \mathbf{H}$. Its characteristic equation uses $(1-\mathrm{i})(1+\mathrm{i}) = 2$:

$$\det(\mathbf{H} - \lambda \mathbf{I}) = (2-\lambda)(3-\lambda) - (1-\mathrm{i})(1+\mathrm{i}) = \lambda^2 - 5\lambda + 4 = (\lambda - 1)(\lambda - 4) = 0,$$

giving the *real* eigenvalues $\lambda = 1$ and $\lambda = 4$, despite the complex entries. Solve $(\mathbf{H} - \lambda \mathbf{I})\mathbf{v} = \mathbf{0}$ for each. For $\lambda = 4$ the first row reads $(2-4)v_1 + (1-\mathrm{i})v_2 = 0$, i.e. $v_1 = \frac{1-\mathrm{i}}{2} v_2$; choosing $v_2 = 2$ clears the fraction and gives $\mathbf{v}_4 = (1-\mathrm{i}, 2)^\mathsf{T}$. The same step for $\lambda = 1$ gives $\mathbf{v}_1$:

$$\lambda = 4: \ \mathbf{v}_4 = \begin{pmatrix} 1-\mathrm{i} \\ 2 \end{pmatrix}, \qquad \lambda = 1: \ \mathbf{v}_1 = \begin{pmatrix} -1+\mathrm{i} \\ 1 \end{pmatrix}.$$

Note (Orthogonality uses a conjugate). For complex vectors, orthogonality is measured by the *conjugate* inner product

$$\mathbf{u}^\dagger \mathbf{v} = \overline{u_1} v_1 + \overline{u_2} v_2,$$

where the conjugate echoes the modulus $|z|^2 = \bar{z}z$ of Lecture 3. The two eigenvectors above are orthogonal in exactly this sense:

$$\mathbf{v}_4^\dagger \mathbf{v}_1 = \overline{(1-\mathrm{i})}(-1+\mathrm{i}) + \overline{2} \cdot 1 = (1+\mathrm{i})(-1+\mathrm{i}) + 2 = -2 + 2 = 0,$$

even though the plain dot product $\mathbf{v}_4^\mathsf{T} \mathbf{v}_1 = 2 + 2\mathrm{i}$ is not zero. The conjugate is essential—a distinction the physics cohort will meet again.

Real eigenvalues are exactly why quantum mechanics represents every measurable quantity by a Hermitian operator: the eigenvalues are the possible measured values. The orthogonality of the eigenvectors—real or complex—has a powerful consequence for diagonalisation, taken up in Section 2.3, and is the foundation of principal-component analysis, where the symmetric covariance matrix of a data set is split into orthogonal directions of greatest variance.

2 Diagonalisation and matrix powers

When a matrix has a full set of eigenvectors they make the best possible coordinate system: in eigenvector coordinates the matrix is simply a list of scale factors. This is *diagonalisation*, and it makes otherwise heavy computations—above all, powers of a matrix—almost free.

2.1 Diagonalisation

Definition 2.1 (Diagonalisable matrix). A square matrix $\mathbf{A}$ is *diagonalisable* if it can be written

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1},$$

with $\mathbf{D}$ diagonal and $\mathbf{P}$ invertible. (Equivalently $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$: the matrices $\mathbf{A}$ and $\mathbf{D}$ are *similar*, describing the same map in different coordinates.)

The eigenvectors and eigenvalues are exactly the ingredients. If $\mathbf{P} = [\,\mathbf{v}_1 \;\cdots\; \mathbf{v}_n\,]$ has the eigenvectors as its columns and $\mathbf{D} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ the eigenvalues, then column by column $\mathbf{AP} = [\,\lambda_1\mathbf{v}_1 \;\cdots\; \lambda_n\mathbf{v}_n\,] = \mathbf{PD}$, so $\mathbf{A} = \mathbf{PDP}^{-1}$ provided $\mathbf{P}$ is invertible.

Theorem 2.1 (The diagonalisation criterion). An $n \times n$ matrix is diagonalisable *if and only if* it has n linearly independent eigenvectors; these form the columns of $\mathbf{P}$ and the corresponding eigenvalues the diagonal of $\mathbf{D}$. In particular, a matrix with *n distinct* eigenvalues is always diagonalisable, since eigenvectors from different eigenvalues are independent.

The last clause is the one part the theorem leaned on without justifying; it is short to prove, and worth proving, since it is what makes distinct eigenvalues enough.

Proposition 2.1 (Distinct eigenvalues give independent eigenvectors). Eigenvectors belonging to *distinct* eigenvalues are linearly independent.

Proof. Take eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ with distinct eigenvalues $\lambda_1 \neq \lambda_2$, and suppose some combination of them vanishes,

$$a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{0}.$$

Apply $\mathbf{A}$ and use $\mathbf{A}\mathbf{v}_i = \lambda_i\mathbf{v}_i$: $a\lambda_1\mathbf{v}_1 + b\lambda_2\mathbf{v}_2 = \mathbf{0}$. Now subtract λ_2 times the first relation, which cancels the $\mathbf{v}_2$ term and leaves

$$a(\lambda_1 - \lambda_2)\mathbf{v}_1 = \mathbf{0}.$$

Since $\mathbf{v}_1 \neq \mathbf{0}$ and $\lambda_1 - \lambda_2 \neq 0$, the coefficient a must be zero; then $b\mathbf{v}_2 = \mathbf{0}$ with $\mathbf{v}_2 \neq \mathbf{0}$ forces $b = 0$ as well. The only combination that vanishes is the trivial one, so $\mathbf{v}_1$ and $\mathbf{v}_2$ are independent. The same step, applied to one eigenvector after another, extends the result to any number of distinct eigenvalues—a short induction completed in the Linear Algebra module. ∎

Remark (Over the reals or over the complex numbers?). When the eigenvalues are complex—as for the rotation of §1.3, with $\lambda = \pm\mathrm{i}$ and eigenvectors $(1, \mp\mathrm{i})$—the matrices $\mathbf{P}$ and $\mathbf{D}$ are complex too: the matrix is diagonalisable *over* $\mathbb{C}$ even though it fixes no real direction and so has no real diagonalisation. The real normal form for such rotations is left to the Linear Algebra module; here "diagonalisable" allows complex $\mathbf{P}$ and $\mathbf{D}$.

Both worked matrices of Section 1 pass this test at once. The 2×2 of Example 1.1 has distinct eigenvalues $2, 3$, and the 3×3 of Example 1.2 has distinct eigenvalues $1, 2, 4$; each therefore has a full set of independent eigenvectors and is diagonalisable. We now put the first of them to work.

2.2 Computing powers

The point of diagonalising is that powers of a diagonal matrix are trivial—just power the entries—and the change of coordinates passes straight through a power:

$$\mathbf{A}^k = (\mathbf{PDP}^{-1})^k = \mathbf{PD}\underbrace{\mathbf{P}^{-1}\mathbf{P}}_{\mathbf{I}}\mathbf{D}\cdots\mathbf{P}^{-1} = \mathbf{PD}^k\mathbf{P}^{-1}, \qquad \mathbf{D}^k = \operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k).$$

A hundredth power of $\mathbf{A}$, which would otherwise need 99 matrix multiplications, costs only two—once $\mathbf{D}^{100}$ is read off by raising the eigenvalues to the power.

Note (Spectral mapping: powers, shifts and inverses keep the eigenvectors). The displayed formula has a one-eigenvector-at-a-time reading: an eigenvector is carried through unchanged, and only its eigenvalue is transformed. If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, applying $\mathbf{A}$ again gives $\mathbf{A}^2\mathbf{v} = \mathbf{A}(\lambda\mathbf{v}) = \lambda^2\mathbf{v}$, and repeating the step,

$$\mathbf{A}^k\mathbf{v} = \lambda^k\mathbf{v}, \qquad (\mathbf{A} + c\mathbf{I})\mathbf{v} = (\lambda + c)\mathbf{v}, \qquad \mathbf{A}^{-1}\mathbf{v} = \tfrac{1}{\lambda}\mathbf{v} \ \ (\lambda \neq 0).$$

The *same* $\mathbf{v}$ serves throughout—this is exactly why $\mathbf{D}^k$ above powers the eigenvalues while $\mathbf{P}$, the matrix of eigenvectors, is left untouched. The inverse case needs $\lambda \neq 0$, which is guaranteed whenever $\mathbf{A}$ is invertible, since $\det\mathbf{A} = \prod_i \lambda_i$. For the $\lambda = 3$ eigenvector $\mathbf{v} = (1,1)^\top$ of Example 1.1, $\mathbf{A}^2\mathbf{v} = 9\mathbf{v}$ and $\mathbf{A}^{-1}\mathbf{v} = \frac{1}{3}\mathbf{v}$ follow at once, with no fresh eigenvector computation.

Example 2.1 ***(A 2×2 diagonalisation used to find $\mathbf{A}^k$).*** Return to $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$, with eigenvalues $2, 3$ and eigenvectors $(2,1)^\top, (1,1)^\top$. Place these as columns and powers on the diagonal:

$$\mathbf{P} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{P}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$$

(using the 2×2 inverse formula, with $\det\mathbf{P} = 1$). Then $\mathbf{A}^k = \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}$; multiplying the first two factors, $\mathbf{P}\mathbf{D}^k = \left(\begin{smallmatrix} 2^{k+1} & 3^k \\ 2^k & 3^k \end{smallmatrix}\right)$, and then by $\mathbf{P}^{-1}$,

$$\mathbf{A}^k = \begin{pmatrix} 2^{k+1} & 3^k \\ 2^k & 3^k \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2^{k+1} - 3^k & 2 \cdot 3^k - 2^{k+1} \\ 2^k - 3^k & 2 \cdot 3^k - 2^k \end{pmatrix}.$$

Setting $k = 1$ recovers $\mathbf{A}$, and $k = 2$ gives $\mathbf{A}^2 = \left(\begin{smallmatrix} -1 & 10 \\ -5 & 14 \end{smallmatrix}\right)$—a closed form for *every* power from a single diagonalisation.

A celebrated application turns the same machinery on the Fibonacci numbers.

Example 2.2 ***(Fibonacci and Binet's formula).*** The Fibonacci sequence obeys $F_{n+1} = F_n + F_{n-1}$, running $0, 1, 1, 2, 3, 5, 8, \ldots$. Stacking two consecutive terms into a vector makes this a single matrix step,

$$\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix}, \qquad \text{so} \qquad \begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = \mathbf{A}^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

The characteristic equation $\lambda^2 - \lambda - 1 = 0$ has the two roots

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618 \quad \text{(the golden ratio)}, \qquad \psi = \frac{1 - \sqrt{5}}{2} \approx -0.618,$$

with eigenvectors $(\varphi, 1)^\top$ and $(\psi, 1)^\top$ and $\varphi - \psi = \sqrt{5}$. Writing the start $(1,0)^\top$ in this eigenbasis gives coefficients $\frac{1}{\sqrt{5}}$ and $-\frac{1}{\sqrt{5}}$, so

$$\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = \frac{1}{\sqrt{5}} \varphi^n \begin{pmatrix} \varphi \\ 1 \end{pmatrix} - \frac{1}{\sqrt{5}} \psi^n \begin{pmatrix} \psi \\ 1 \end{pmatrix}.$$

Reading off the second component gives *Binet's formula*, a closed form for the nth Fibonacci number out of nothing but eigenvalues:

$$\boxed{F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}}.$$

Because $|\psi| < 1$, the ψ^n term vanishes and F_n is simply the nearest integer to $\varphi^n/\sqrt{5}$; the ratio F_{n+1}/F_n tends to φ. The *larger* eigenvalue sets the growth—a first glimpse of the dominant-eigenvalue idea that drives Section 3.

2.3 Symmetric matrices and orthogonal diagonalisation

For a symmetric matrix the eigenvectors are not merely independent but *orthogonal* (§1.4), and that makes its diagonalisation especially clean. Scaling each eigenvector to unit length gives an *orthonormal* set, so the eigenvector matrix becomes an *orthogonal* matrix $\mathbf{Q}$, one whose inverse is simply its transpose, $\mathbf{Q}^{-1} = \mathbf{Q}^\mathsf{T}$.

Theorem 2.2 (Spectral theorem, real symmetric case). Every real symmetric matrix can be written

$$\mathbf{A} = \mathbf{QDQ}^\mathsf{T},$$

with $\mathbf{D}$ diagonal (the real eigenvalues) and $\mathbf{Q}$ orthogonal (orthonormal eigenvectors as columns). In particular a symmetric matrix is always diagonalisable. (Eigenvectors from *distinct* eigenvalues are automatically orthogonal; within a repeated eigenvalue's eigenspace an orthonormal basis is *chosen*, e.g. by Gram–Schmidt.)

Example 2.3 ***(Orthogonal diagonalisation of the*** 3×3***).*** The symmetric matrix $\mathbf{A} = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{smallmatrix}\right)$ has eigenvalues $1, 2, 4$ and the mutually orthogonal eigenvectors found earlier. Their lengths are $\sqrt{3}$, $\sqrt{2}$, $\sqrt{6}$, so the normalised columns

$$\mathbf{Q} = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ -1/\sqrt{3} & 0 & 2/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \end{pmatrix}, \qquad \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

satisfy $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}$ and $\mathbf{A} = \mathbf{QDQ}^\mathsf{T}$. This decomposition into perpendicular axes, each with its own scale factor, is exactly the engine of principal-component analysis: the orthonormal eigenvectors of a covariance matrix are the principal directions of the data, and the eigenvalues measure the spread along each.

2.4 When diagonalisation fails

Not every matrix is diagonalisable. The obstruction is always a repeated eigenvalue that fails to supply enough eigenvectors.

Example 2.4 ***(A defective matrix).*** The matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ has characteristic equation

$(2-\lambda)^2 = 0$, so $\lambda = 2$ is a *double* root. But solving $(\mathbf{A} - 2\mathbf{I})\mathbf{v} = \mathbf{0}$,

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \mathbf{v} = \mathbf{0} \implies v_2 = 0,$$

leaves only the single eigen-direction $(1,0)^{\mathsf{T}}$. There is one independent eigenvector where two are needed, so $\mathbf{A}$ cannot be diagonalised.

Definition 2.2 (Algebraic and geometric multiplicity). For an eigenvalue λ, its *algebraic multiplicity* is the number of times it repeats as a root of the characteristic polynomial, and its *geometric multiplicity* is the dimension of its eigenspace—the number of independent eigenvectors it provides. The geometric multiplicity never exceeds the algebraic one, and

$$\mathbf{A} \text{ is diagonalisable} \iff \text{geometric} = \text{algebraic for } \textit{every} \text{ eigenvalue.}$$

In the defective example the eigenvalue 2 has algebraic multiplicity 2 but geometric multiplicity 1, and the shortfall is exactly what blocks diagonalisation. A repeated eigenvalue is therefore *harmless* when its eigenspace is as large as its repeat count—the symmetric matrices of Section 2.3, for instance, are never defective—and distinct eigenvalues avoid the question entirely, each forcing one eigenvector. When an eigenspace does fall short, the missing directions are filled by *generalised* eigenvectors, and the nearest-to-diagonal form a matrix can then reach is the *Jordan canonical form*—both developed in the Linear Algebra module.

3 Application: dynamical systems and Markov chains

Eigenvalues come into their own when a system evolves by repeated multiplication. If a state vector updates each step by $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$, then after n steps $\mathbf{x}_n = \mathbf{A}^n\mathbf{x}_0$—and the diagonalisation of Section 2 turns the long-run behaviour into a question about the eigenvalues alone.

3.1 Markov chains

A *Markov chain* models a system that moves between a fixed set of states with given probabilities. The state vector $\mathbf{x}_n$ holds the probability of being in each state, and a *transition matrix* $\mathbf{P}$—whose columns are probabilities and so sum to one—advances it one step by $\mathbf{x}_{n+1} = \mathbf{P}\mathbf{x}_n$.

Example 3.1 *(A two-state chain)*. Customers choose between brands A and B. Each year an A-customer stays with probability 0.8 and defects to B with 0.2; a B-customer stays with 0.7 and switches to A with 0.3 (Figure 2). With $\mathbf{x}_n = (\text{fraction with } A,\ \text{fraction with } B)$, the transition matrix is

$$\mathbf{P} = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix}, \qquad \mathbf{x}_{n+1} = \mathbf{P}\mathbf{x}_n.$$

Starting from an even split $\mathbf{x}_0 = (0.5, 0.5)$, one step gives $\mathbf{x}_1 = \mathbf{P}\mathbf{x}_0 = (0.55,\ 0.45)$. Where does the market settle?

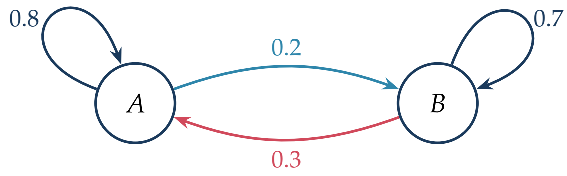

Figure 2: The two-state chain: arrows carry the one-step transition probabilities, and the probabilities leaving each state sum to one.

3.2 The steady state is an eigenvector

A distribution that no longer changes satisfies $\mathbf{P}\mathbf{x} = \mathbf{x}$—an eigenvector with eigenvalue 1. Every transition matrix has this eigenvalue, and finding its eigenvector finds the equilibrium.

Example 3.2 ***(The long-run distribution).*** The characteristic equation of $\mathbf{P}$ is

$$\det\begin{pmatrix} 0.8-\lambda & 0.3 \\ 0.2 & 0.7-\lambda \end{pmatrix} = \lambda^2 - 1.5\lambda + 0.5 = (\lambda-1)(\lambda-0.5) = 0,$$

so $\lambda_1 = 1$ and $\lambda_2 = 0.5$. For $\lambda_1 = 1$, $(\mathbf{P}-\mathbf{I})\mathbf{x} = \mathbf{0}$ gives $-0.2x_1 + 0.3x_2 = 0$, i.e. $x_1 = \frac{3}{2}x_2$, so the steady-state direction is $(3, 2)$; scaled to a probability distribution (entries summing to one) it is

$$\mathbf{x}_\infty = \big(0.6,\ 0.4\big).$$

Indeed $\mathbf{P}(0.6, 0.4) = (0.6, 0.4)$: in the long run 60% hold brand A and 40% brand B, whatever the starting split.

3.3 Why this chain converges

Diagonalisation explains the convergence. Writing $\mathbf{x}_0 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ in the eigenvector basis,

$$\mathbf{x}_n = \mathbf{P}^n\mathbf{x}_0 = c_1\,\lambda_1^n\mathbf{v}_1 + c_2\,\lambda_2^n\mathbf{v}_2 = c_1\,\mathbf{v}_1 + c_2\,(0.5)^n\,\mathbf{v}_2.$$

The first term, with eigenvalue 1, stands still; the second, with eigenvalue 0.5, decays to nothing. So $\mathbf{x}_n$ slides to the steady state $c_1\mathbf{v}_1$ whatever the starting split (Figure 3).

Remark (When does a Markov chain settle?). The decay above worked because the *second* eigenvalue had modulus below 1. In general a Markov chain settles to a unique steady state, independent of the start, if every eigenvalue other than the $\lambda = 1$ is smaller than 1 in modulus (a *regular* chain). A chain with a second eigenvalue of modulus 1 need not converge: $\mathbf{P} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has eigenvalues ± 1 and simply oscillates forever. The full statement is the Perron–Frobenius theorem of the Linear Algebra module. More broadly, for a general (non-stochastic) system $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$ the *dominant* eigenvalue—the one of largest modulus—decides the fate: modulus greater than one gives unbounded growth (as for the Fibonacci matrix, with $\varphi > 1$), and less than one gives collapse to zero.

Example 3.3 ***(The evolution in closed form).*** The second eigenvalue $\lambda_2 = 0.5$ has eigenvector

$(1,-1)$, from $(\mathbf{P}-0.5\mathbf{I})\mathbf{x}=\mathbf{0}$. Decomposing the even-split start in the eigenvector basis,

$$\mathbf{x}_0 = (0.5, 0.5) = \underbrace{1}_{c_1} \cdot (0.6, 0.4) + \underbrace{(-0.1)}_{c_2} \cdot (1, -1),$$

so $c_1 = 1$ and $c_2 = -0.1$. (Here $c_1 = 1$ for *any* probability start, since $(0.6, 0.4)$ already sums to one while $(1, -1)$ sums to zero.) Propagating each eigenvector by its eigenvalue,

$$\mathbf{x}_n = (0.6, 0.4) + (-0.1)(0.5)^n(1, -1),$$

so the fraction holding brand A is $0.6 - 0.1(0.5)^n$. It rises $0.5, 0.55, 0.575, 0.5875, \ldots$ towards 0.6, halving its remaining gap each year—convergence at exactly the rate of the second eigenvalue, as the next figure shows.

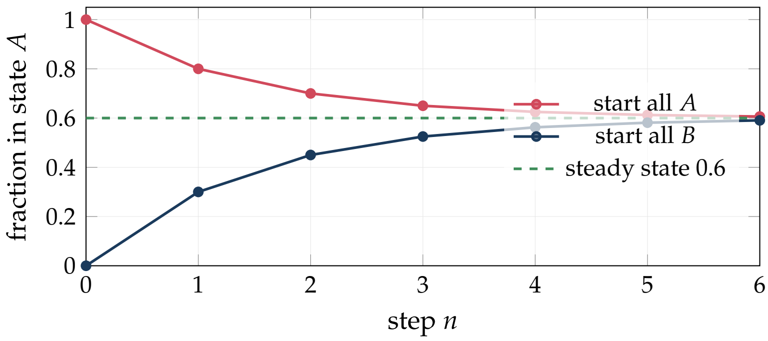

Figure 3: Two very different starts both converge to the steady state 0.6, at the rate set by the second eigenvalue 0.5.

Remark (The same idea, far afield). This dominant-eigenvalue principle is everywhere. Google's original PageRank ranks web pages by the steady-state eigenvector of a giant transition matrix of links; a population's long-term age profile is the dominant eigenvector of its growth matrix; and the stability of an equilibrium in physics or economics is decided by whether the governing eigenvalues have modulus below one. The eigenvector is the destination, and the dominant eigenvalue is the speed.

Summary

Concept	Key idea
Eigenpair	$\mathbf{Av} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$: a direction the matrix only scales, by λ.
Eigenvalues	Roots of the characteristic equation $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$ (degree n).
Eigenvectors	Non-zero solutions of $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ (Gaussian elimination), one eigenspace per eigenvalue.
Trace/det check	$\sum \lambda_i = \operatorname{tr}\mathbf{A}$ and $\prod \lambda_i = \det\mathbf{A}$.
Complex eigenvalues	Signal rotation; a plane rotation has $\lambda = e^{\pm i\theta}$, modulus the scaling, argument the frequency.
Symmetric/Hermitian	$\mathbf{A}^\mathsf{T} = \mathbf{A}$ (or $\mathbf{A}^\dagger = \mathbf{A}$): real eigenvalues, orthogonal eigenvectors; never defective.
Diagonalisation	$\mathbf{A} = \mathbf{PDP}^{-1}$, $\mathbf{P}$'s columns the eigenvectors, $\mathbf{D}$ the eigenvalues. Possible iff n independent eigenvectors (distinct eigenvalues suffice).
Spectral theorem	Symmetric $\mathbf{A} = \mathbf{QDQ}^\mathsf{T}$ with $\mathbf{Q}$ orthogonal ($\mathbf{Q}^{-1} = \mathbf{Q}^\mathsf{T}$); basis of PCA.
Matrix powers	$\mathbf{A}^k = \mathbf{PD}^k\mathbf{P}^{-1}$ with $\mathbf{D}^k = \operatorname{diag}(\lambda_i^k)$. Fibonacci $\to$ Binet $F_n = (\varphi^n - \psi^n)/\sqrt{5}$.
Multiplicity	Diagonalisable iff geometric $=$ algebraic for every eigenvalue; otherwise *defective* (e.g. $\left(\begin{smallmatrix}2&1\\0&2\end{smallmatrix}\right)$).
Markov chain	$\mathbf{x}_{n+1} = \mathbf{P}\mathbf{x}_n$, columns of $\mathbf{P}$ summing to one; steady state is the $\lambda = 1$ eigenvector.
Long run	$\mathbf{x}_n = \mathbf{P}^n\mathbf{x}_0$; the *dominant* eigenvalue decides stability/growth/decay, its eigenvector the destination.

Looking back, and forward. This completes the linear-algebra *strand: from vectors and matrices to the eigenvalues and eigenvectors that expose the directions a matrix merely stretches. These ideas return—for mathematics and data-science students—in a dedicated linear-algebra course, where the eigenvalues reappear as the spectral theorem, generalised eigenvectors and the singular value decomposition; and for physicists the same eigenvalues resurface as normal modes and quantum states. The module now changes subject for its second half—the* differential and integral calculus *of one variable—beginning next time with the derivative.*

Companion material: a separate *Exercise Sheet 6* (with solutions) develops the practice problems for this unit.